

CARTERSVILLE, GA, USA

WMO: 722156

Lat: 34.123N

Lon: 84.849W

Elev: 233

StdP: 98.56

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53873

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.9	(c) -4.8	(d) -14.7	(e) 1.1	(f) -2.8	(g) -12.5	(h) 1.3	(i) -1.0	(j) 6.9	(k) 8.1	(l) 5.7	(m) 7.9	(n) 1.1	(o) 340	(p) 0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.1	(c) 34.5	(d) 23.3	(e) 33.3	(f) 23.3	(g) 32.2	(h) 23.2	(i) 25.5	(j) 30.8	(k) 25.0	(l) 30.3	(m) 24.5	(n) 29.6	(o) 2.0	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.5	27.3	23.5	18.8	26.9	23.0	18.2	26.4	79.0	30.7	76.9	30.4	75.1	29.6	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.6	(b) 4.9	(c) 4.3	(e) -10.3	(f) 36.2	(g) 2.8	(h) 1.9	(i) -12.3	(j) 37.5	(k) -13.9	(l) 38.6	(m) -15.5	(n) 39.7	(o) -17.5	(p) 41.0		
(a) 5.6	(b) 4.9	(c) 4.3	(e) -11.2	(f) 26.7	(g) 2.6	(h) 0.7	(i) -13.1	(j) 27.2	(k) -14.6	(l) 27.6	(m) -16.1	(n) 28.0	(o) -18.0	(p) 28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	5.4	7.4	11.7	16.0	20.4	24.5	25.8	25.9	22.8	16.7	10.8	7.3
	DBStd	8.30	5.53	4.90	4.79	3.85	3.50	2.14	1.74	1.98	3.02	4.41	4.53	4.88
	HDD10.0	470	163	101	39	4	0	0	0	0	0	4	43	116
	HDD18.3	1685	400	307	211	87	23	0	0	0	4	82	229	342
	CDD10.0	2761	22	28	90	186	322	436	490	492	384	213	67	32
	CDD18.3	934	0	1	4	18	86	186	232	234	139	32	3	1
	CDH23.3	8374	1	4	53	249	813	1691	2035	2087	1159	265	16	1
	CDH26.7	3413	0	0	4	41	258	734	892	957	468	58	1	0
Wind	WSAvg	1.3	1.5	1.5	1.7	1.5	1.2	1.0	0.9	1.0	1.1	1.2	1.3	1.4
Precipitation	PrecAvg	1283	122	117	143	99	97	100	122	96	97	85	103	104
	PrecMax	1746	197	231	244	193	260	171	273	220	276	228	200	214
	PrecMin	712	58	48	51	42	26	41	19	34	10	8	40	34
	PrecStd	221	37	41	56	36	52	40	55	42	69	55	44	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.8	23.5	27.2	29.7	32.8	35.5	35.1	36.7	35.3	30.6	25.8	22.5
		MCWB	15.9	17.6	17.6	19.0	21.7	22.8	23.8	23.5	22.3	21.4	18.5	17.4
	2%	DB	19.1	20.5	24.8	27.8	31.0	33.7	33.8	34.5	33.0	28.0	22.8	20.1
		MCWB	14.6	15.2	16.5	18.3	21.1	22.9	24.1	23.6	21.8	19.9	16.6	16.8
	5%	DB	17.1	18.6	22.7	26.4	29.5	32.4	32.5	33.0	31.3	26.3	21.1	17.9
		MCWB	13.6	14.0	15.3	17.5	20.6	22.9	24.2	23.6	21.9	18.8	15.8	14.7
	10%	DB	14.7	16.3	20.4	24.4	27.9	31.0	31.3	31.7	29.5	24.2	19.0	15.6
		MCWB	12.0	11.4	14.0	16.7	20.4	22.6	23.9	23.4	21.7	18.0	14.8	12.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	18.5	18.9	21.2	23.7	25.6	26.4	26.1	24.5	23.6	21.2	19.5
		MCDB	20.0	21.8	24.3	26.8	28.0	30.6	31.2	31.5	29.6	27.6	22.6	20.9
	2%	WB	15.9	17.1	17.9	20.1	22.8	24.9	25.6	25.3	23.9	22.1	18.9	17.4
		MCDB	18.1	19.6	22.6	25.3	28.0	30.2	30.7	30.9	29.0	25.8	21.0	19.3
	5%	WB	14.4	15.1	16.8	19.0	22.0	24.3	25.1	24.8	23.2	20.9	17.3	15.5
		MCDB	16.7	17.4	20.9	23.9	27.2	29.5	30.3	30.2	28.1	24.0	19.6	17.4
	10%	WB	12.2	12.9	15.4	18.0	21.3	23.7	24.6	24.3	22.5	19.6	15.5	13.3
		MCDB	14.2	15.3	19.1	22.4	26.2	28.6	29.5	29.3	27.0	22.9	18.2	15.1
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.6	12.8	13.7	12.7	12.0	11.1	11.3	12.0	12.8	13.0	10.7
		MCDBR	13.9	14.3	15.7	16.5	14.9	14.3	12.7	13.6	14.6	14.5	15.0	12.7
		MCWBR	9.5	9.3	8.4	7.7	6.2	5.3	4.7	4.5	5.1	6.8	8.8	9.2
	5% WB	MCDBR	11.2	11.1	12.6	12.9	12.0	11.4	11.2	11.0	11.1	10.6	10.6	10.4
MCWBR		8.5	8.6	7.4	6.7	5.6	4.9	4.7	4.2	4.5	5.8	7.6	8.5	
Clear-Sky Solar Irradiance	taub	0.317	0.319	0.355	0.394	0.457	0.485	0.534	0.532	0.425	0.373	0.343	0.318	
	taud	2.514	2.502	2.427	2.307	2.167	2.130	2.009	1.999	2.288	2.429	2.449	2.529	
	Ebn at Noon	896	932	919	893	837	810	768	761	836	859	853	873	
	Edh at Noon	83	94	109	129	150	156	175	174	124	100	88	78	
All-Sky Solar Radiation	RadAvg	2.60	3.14	4.25	5.42	5.98	6.17	5.88	5.47	4.72	3.95	2.91	2.20	
	RadStd	0.17	0.40	0.36	0.35	0.48	0.48	0.42	0.31	0.46	0.50	0.32	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+129
(47) Regional (40 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+152	+108

Nomenclature: See separate page